EDMD for piecewise expanding analytic interval maps using Legendre polynomials

Abstract. We consider EDMD for piecewise expanding analytic interval maps using Legendre polynomials.

1. Basic estimates

Recall that the Legendre polynomials P_n are given by the following generating function

$$\sum_{n=0}^{\infty} P_n(x)t^n = \frac{1}{\sqrt{1-2xt+t^2}}\,. \tag{1}$$

We first collect a few properties of the Legendre polynomials considered as functions on $[-1,1]$. Setting $x=1$ in above identity we find that

$$\sum_{n=0}^{\infty} P_n(1)t^n = \frac{1}{1-t}$$

from which

$$P_n(1) = 1 \quad (n \in \mathbb{N}_0)\,, \tag{2}$$

while differentiating (1) with respect to x followed by evaluation at $x=1$ gives

$$\sum_{n=0}^{\infty} P_n'(1)t^n = \frac{t}{(1-t)^3}\,,$$

which in turn yields

$$P_n'(1) = \frac{n(n+1)}{2} \quad (n \in \mathbb{N}_0)\,. \tag{3}$$

Slightly deeper is the following result, which underpins a number of the estimates to follow.

1.1. Lemma. *Given $z \in \mathbb{C}$ write $w = \frac{1}{2}(z+z^{-1})$. Then*

$$P_n(w) = \sum_{k=0}^{n} \alpha_k \alpha_{n-k} z^{n-2k} \quad (n \in \mathbb{N}_0)\,,$$

where

$$\alpha_0 = 1, \quad \alpha_k = \frac{1\cdot 3\cdots(2k-1)}{2\cdot 4\cdots 2k} = \frac{(2k)!}{2^{2k}(k!)^2} \quad (k \in \mathbb{N})\,.$$

Proof. See [**1**, Lemma 12.4.1]. □

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A simple consequence of the lemma above is the following well-known result, the simple proof of which we provide for the benefit of the reader.

1.2. Lemma. *We have*

$$\sup_{x\in[-1,1]} |P_n(x)| = 1 \quad (n \in \mathbb{N}_0)\,.$$

Proof. By Lemma 1.1 we have for $n \in \mathbb{N}_0$

$$P_n(\cos(t)) = \sum_{k=0}^{n} \alpha_k \alpha_{n-k} \exp(i(n-2k)t)\,,$$

and so for $t \in \mathbb{R}$

$$|P_n(\cos(t))| \le \sum_{k=0}^{n} \alpha_k \alpha_{n-k} = P_n(\cos(0)) = 1\,,$$

by (2), and the assertion follows. □

1.3. Lemma. *We have*

$$\sup_{x\in[-1,1]} |P_n'(x)| = \frac{n(n+1)}{2} \quad (n \in \mathbb{N}_0)\,.$$

Proof. We start by observing that the derivatives of the Legendre polynomials satisfy the following recurrence relation

$$P_{n+1}'(x) = (n+1)P_n(x) + xP_n'(x) \quad (n \in \mathbb{N}_0)\,. \tag{4}$$

The relation follows from the fact that the generating function G of the Legendre polynomials, that is,

$$\sum_{n=0}^{\infty} P_n(x)t^n = \frac{1}{\sqrt{1-2xt+t^2}} = G(x,t)$$

satisfies

$$\frac{1}{t}\frac{\partial}{\partial x}G(x,t) = \frac{\partial}{\partial t}\left(tG(x,t)\right) + x\frac{\partial}{\partial x}G(x,t)\,,$$

which is easily verified. For $n \in \mathbb{N}_0$, let now

$$M_n = \sup_{x\in[-1,1]} |P_n'(x)|\,.$$

Using the recurrence relation (4) we have for $x \in [-1,1]$

$$\left|P_{n+1}'(x)\right| = (n+1)\,|P_n(x)| + |x|\;|P_n'(x)| \le (n+1) + M_n$$

by Lemma 1.2, and so

$$M_{n+1} \le (n+1) + M_n \quad (n \in \mathbb{N}_0)\,.$$

Since $M_0 = 0$, the inequality above gives

$$M_n \le \sum_{k=0}^{n} k = \frac{n(n+1)}{2}\,.$$

But, by (3), we have

$$M_n \ge P_n'(1) = \frac{n(n+1)}{2}\,,$$

and so

$$M_n = \frac{n(n+1)}{2}$$

as claimed. □

We now turn to properties of the Legendre polynomials in the complex plane. Another look at Lemma 1.1 suggests that, just as with Chebyshev polynomials, elliptic regions play a useful role for analysing convergence properties of Legendre polynomials for analytic functions.

1.4. Notation. Given $r \in \mathbb{R}$ with $r > 1$ we write

$$E_r = \left\{ \frac{1}{2}(z + z^{-1}) \,:\, z \in \mathbb{C},\, 1 \le |w| < r \right\},$$

for the interior of the ellipse with the lengths of the major and minor semi-axes given by $\frac{1}{2}(r + r^{-1})$ and $\frac{1}{2}(r - r^{-1})$, respectively, and foci at -1 and 1.

The positively oriented boundary of E_r will be parametrised by

$$\gamma_r : [0, 2\pi] \to \mathbb{C}, \quad \gamma_r(t) = \frac{1}{2}\left(r \exp(it) + r^{-1} \exp(-it)\right).$$

With these elliptic regions (also known as 'Bernstein ellipses' in numerical analysis) we associate the following Hardy spaces.

1.5. Definition. Let $r > 1$.

(i) $A(E_r)$ is the Banach space of functions holomorphic on E_r and continuous on the closure of E_r equipped with the norm

$$\|f\|_{A(E_r)} = \sup\{|f(z)| \,:\, z \in \mathrm{clo}(E_r)\}.$$

(ii) $H^\infty(E_r)$ is the Banach space of bounded holomorphic functions on E_r equipped with the norm

$$\|f\|_{H^\infty(E_r)} = \sup\{|f(z)| \,:\, z \in E_r)\}.$$

(iii) $H^2(E_r)$ is the Hilbert space obtained as the completion of $A(E_r)$ with respect to the inner product

$$(f, g)_{H^2(E_r)} = \frac{1}{2\pi} \int_0^{2\pi} f(\gamma_r(t)) \overline{g(\gamma_r(t))}\, dt.$$

It is not difficult to see that the canonical embeddings

$$A(E_r) \hookrightarrow H^\infty(E_r) \hookrightarrow H^2(E_r)$$

are continuous. Clearly, the Legendre polynomials belong to each of these spaces. Moreover, we have the following bounds for their norms.

1.6. Lemma. *Let $r > 1$. Then*

(a) $\|P_n\|_{A(E_r)} = \|P_n\|_{H^\infty(E_r)} \le \frac{1}{2}(r^n + r^{-n})$ *for any* $n \in \mathbb{N}_0$;

(b) $\|P_n\|_{H^2(E_r)} \le \sqrt{\frac{1}{2}(r^{2n} + r^{-2n})}$ *for any* $n \in \mathbb{N}_0$.

Proof.

(a) Fix $n \in \mathbb{N}_0$. By the maximum modulus principle there is $z \in \mathbb{C}$ with $|z| = r$ such that

$$\|P_n\|_{A(E_r)} = \|P_n\|_{H^\infty(E_r)} = |P_n(w)|$$

where $w = \frac{1}{2}(z + z^{-1})$. By Lemma 1.1 we have

$$|P_n(w)| \le \sum_{k=0}^{n} \alpha_k \alpha_{n-k} r^{n-2k}.$$

We now distinguish two cases. If n is odd then

$$\sum_{k=0}^{n} \alpha_k \alpha_{n-k} r^{n-2k} = \sum_{k=0}^{\frac{n-1}{2}} \alpha_k \alpha_{n-k} \left(r^{n-2k} + r^{-(n-2k)} \right)$$
$$\leq \frac{1}{2} \left(r^n + r^{-n} \right) \sum_{k=0}^{\frac{n-1}{2}} 2\alpha_k \alpha_{n-k} \,.$$

But

$$\sum_{k=0}^{\frac{n-1}{2}} 2\alpha_k \alpha_{n-k} = \sum_{k=0}^{n} \alpha_k \alpha_{n-k} = P_n(1) = 1$$

and the assertion follows in this case.

If, on the other hand, n is even then

$$\sum_{k=0}^{n} \alpha_k \alpha_{n-k} r^{n-2k} = \sum_{k=0}^{\frac{n}{2}-1} \alpha_k \alpha_{n-k} \left(r^{n-2k} + r^{-(n-2k)} \right) + \alpha_{\frac{n}{2}}^2$$
$$\leq \frac{1}{2} \left(r^n + r^{-n} \right) \left(\sum_{k=0}^{\frac{n}{2}-1} 2\alpha_k \alpha_{n-k} + \alpha_{\frac{n}{2}}^2 \right) .$$

But

$$\sum_{k=0}^{\frac{n}{2}-1} 2\alpha_k \alpha_{n-k} + \alpha_{\frac{n}{2}}^2 = \sum_{k=0}^{n} \alpha_k \alpha_{n-k} = P_n(1) = 1$$

and the assertion follows in this case as well.

(b) Fix $n \in \mathbb{N}_0$. Then, using Lemma 1.1 again, we have

$$\|P_n\|_{H^2(E_r)}^2 = \frac{1}{2\pi} \int_0^{2\pi} |P_n(\gamma_r(t))|^2 \, dt$$
$$= \frac{1}{2\pi} \int_0^{2\pi} \sum_{k,l=0}^{n} \alpha_k \alpha_{n-k} \alpha_l \alpha_{n-l} r^{2n-2(k+l)} \exp(2i(l-k)t) \, dt$$
$$= \sum_{k=0}^{n} \alpha_k^2 \alpha_{n-k}^2 r^{2(n-2k)} \,.$$

By distinguishing the case of n odd or even as in the proof of part (a), it is not difficult to see that

$$\sum_{k=0}^{n} \alpha_k^2 \alpha_{n-k}^2 r^{2(n-2k)} \leq \frac{1}{2}(r^{2n} + r^{-2n}) \sum_{k=0}^{n} \alpha_k^2 \alpha_{n-k}^2 \leq \frac{1}{2}(r^{2n} + r^{-2n}) \,,$$

and the assertion follows. □

As is well known, the Legendre polynomials are orthogonal in $L^2[-1,1]$ with

$$\int_{-1}^{1} P_n(x) P_m(x) \, dx = \frac{2}{2n+1} \delta_{m,n} \quad (m, n \in \mathbb{N}_0) \,.$$

Thus, if f is a polynomial of degree n we have

$$f = \sum_{k=0}^{n} a_k(f) P_k \tag{5}$$

where

$$a_n(f) = (n + \frac{1}{2}) \int_{-1}^{1} f(x) P_n(x)\, dx\,.$$

Studying the expansion (5) for more general classes of functions has a comparatively long history going back to the 19th century. In the following we are interested in the asymptotic behaviour of the coefficient functionals a_n as $n \to \infty$ for functions that are analytic. Estimates of this type again have a considerable pedigree going back to C. Neumann (see [**1**, Theorem 12.4.7]). For a more recent account, see [**2**], where estimates can be found for functions in $A(E_r)$. As we are interested in estimates for functions in the Hardy spaces $H^\infty(E_r)$ and $H^2(E_r)$ we shall revisit the argument in [**2**, Theorem 2.1], which in turn is a refinement of the classical argument in [**1**, Theorem 12.4.7]).

In order to proceed, suppose that $n \in \mathbb{N}_0$ and let Q_n denote the Legendre function of the second kind given by

$$Q_n(z) = \frac{1}{2} \int_{-1}^{1} \frac{P_n(x)}{z - x}\, dt \quad (z \in \mathbb{C} \setminus [-1, 1]))\,.$$

The reason these functions appear in this context is due to the following simple observation.

1.7. Lemma. *Let $r > 1$. Given $f \in A(E_r)$ we have for any $n \in \mathbb{N}_0$*

$$a_n(f) = \frac{2n+1}{2\pi i} \int_{\partial E_r} f(w) Q_n(w)\, dw\,,$$

where ∂E_r denotes the positively oriented boundary of E_r.

Proof. Given $f \in A(E_r)$ we can write

$$f(x) = \frac{1}{2\pi i} \int_{\partial E_r} \frac{f(w)}{w - x}\, dw \quad (x \in [-1, 1])\,.$$

Thus, by Fubini's theorem, we have

$$\begin{aligned} a_n(f) &= (n + \frac{1}{2}) \int_{-1}^{1} f(x) P_n(x)\, dx \\ &= (n + \frac{1}{2}) \int_{-1}^{1} \frac{1}{2\pi i} \int_{\partial E_r} \frac{f(w)}{w - x}\, dw\, P_n(x)\, dx \\ &= \frac{2n+1}{2\pi i} \int_{\partial E_r} f(w)\, \frac{1}{2} \int_{-1}^{1} \frac{P_n(x)}{w - x}\, dx\, dw \\ &= \frac{2n+1}{2\pi i} \int_{\partial E_r} f(w) Q_n(w)\, dw\,. \end{aligned}$$

□

The following is a summary of the main ingredients from the proof of Theorem 2.1 in [**2**].

1.8. Lemma. *Let $n \in \mathbb{N}_0$. For $w = \frac{1}{2}(z + z^{-1})$ with $|z| > 1$ we have*

$$Q_n(w) = \sum_{k=n+1}^{\infty} \frac{\sigma_{n,k}}{z^k}\,, \tag{6}$$

where

$$\sigma_{n,k} = \int_0^\pi P_n(\cos(t)) \sin(kt)\, dt \quad (n, k \in \mathbb{N}_0)\,. \tag{7}$$

Moreover, for $n \in \mathbb{N}_0$ we have

$$\sigma_{n,n+2k} = 0 \quad (k \geq 1) \tag{8}$$

and

$$0 \leq \sigma_{n,k} \leq 2 \quad (k \geq n+1)\,. \tag{9}$$

PROOF. For the expansion (6), see [**1**, Lemma 12.4.6], while (8) and (9) follow from Equations (2.5) and (2.6) in [**2**], respectively. In fact, (8) follows from parity considerations, by writing $\sigma_{n,k}$ using the Chebyshev polynomials of the second kind. As for the upper bound in (9), this can also be seen more easily by using Lemma 1.2 and the fact that

$$\int_0^\pi |\sin(kt)|\ dt = 2 \quad (k \in \mathbb{N})\,.$$

The lower bound in (9) in turn can be deduced from Lemma 1.1 and the fact that

$$\int_0^\pi \cos(lt)\sin(kt)\, dt = \frac{1}{2}\left(1 - (-1)^{l+k}\right) \frac{2k}{k^2 - l^2} \geq 0 \quad (k, l \in \mathbb{N}_0, k > l)\,.$$

□

The following estimate looks superficially the same as (2.2) from [**2**, Theorem 2.1]. Note, however, that we only assume that f is analytic in E_r (and bounded there), rather than that f is analytic in and on E_r.

1.9. LEMMA. *Let $r > 1$ and $f \in H^\infty(E_r)$. Then for every $n \in \mathbb{N}_0$ we have*

$$|a_n(f)| \leq (2n+1) \frac{|\partial E_r|}{\pi(r - r^{-1})} \|f\|_{H^\infty(E_r)}\, r^{-n}\,,$$

where $|\partial E_r|$ denotes the length of the ellipse ∂E_r.

PROOF. Fix ρ with $1 < \rho < r$. By hypothesis $f \in H^\infty(E_r)$, and so $f \in A(E_\rho)$. Using Lemma 1.7 we have

$$\begin{aligned} |a_n(f)| &\leq \frac{2n+1}{2\pi} \int_{\partial E_\rho} |f(w)|\, |Q_n(w)|\ |dw| \\ &\leq \frac{2n+1}{2\pi} |\partial E_\rho|\, \|f\|_{A(E_\rho)} \sup_{w \in \partial E_\rho} |Q_n(w)| \\ &\leq \frac{2n+1}{2\pi} |\partial E_r|\, \|f\|_{H^\infty(E_r)} \sup_{w \in \partial E_\rho} |Q_n(w)|\ . \end{aligned}$$

But by Lemma 1.8 we have for $w \in \partial E_\rho$

$$|Q_n(w)| \leq \sum_{k=0}^\infty \frac{|\sigma_{n,n+2k+1}|}{\rho^{n+2k+1}} \leq \frac{2}{1 - \rho^{-2}} \rho^{-(n+1)}\,,$$

and thus

$$|a_n(f)| \leq \frac{2n+1}{2\pi} |\partial E_r|\, \|f\|_{H^\infty(E_r)} \frac{2}{\rho - \rho^{-1}} \rho^{-n}\,.$$

But since $\rho < r$ was arbitrary, and $\rho \mapsto (\rho - \rho^{-1})^{-1}\rho^{-n}$ is monotonically decreasing for $\rho > 1$, the assertion follows. □

1.10. Lemma. *Let $r > 1$ and $f \in H^2(E_r)$. Then for every $n \in \mathbb{N}_0$ we have*

$$|a_n(f)| \leq (2n+1)\sqrt{\frac{r^2+r^{-2}}{r^2-r^{-2}}}\,\|f\|_{H^2(E_r)}\,r^{-n}\,.$$

Proof. Since $A(E_r)$ is, by construction, dense in $H^2(E_r)$, we may with no loss of generality assume that $f \in A(E_r)$. Using Lemma 1.7 and the Cauchy–Schwarz inequality we have

$$\begin{aligned}
|a_n(f)|^2 &= (2n+1)^2\left|\frac{1}{2\pi}\int_0^{2\pi} f(\gamma_r(t))\,Q_n(\gamma_r(t))\,\gamma_r'(t)\,dt\right|^2\\
&\leq (2n+1)^2\frac{1}{2\pi}\int_0^{2\pi}|f(\gamma_r(t))|^2\,dt\,\frac{1}{2\pi}\int_0^{2\pi}|Q_n(\gamma_r(t))|^2\,|\gamma_r'(t)|^2\,dt\\
&= (2n+1)^2\,\|f\|^2_{H^2(E_r)}\,q_n
\end{aligned}$$

with

$$q_n = \int_0^{2\pi}|Q_n(\gamma_r(t))|^2\,|\gamma_r'(t)|^2\,dt\,.$$

But by Lemma 1.8 we have

$$\begin{aligned}
q_n &= \frac{1}{2\pi}\int_0^{2\pi}\sum_{k,l=n+1}^{\infty}\frac{\sigma_{n,k}\sigma_{n,l}}{r^{k+l}}\exp(i(l-k)t)\frac{r^2+r^{-2}-\exp(2it)-\exp(-2it)}{4}\,dt\\
&= \frac{r^2+r^{-2}}{4}\sum_{k=n+1}^{\infty}\frac{\sigma_{n,k}^2}{r^{2k}} - \frac{1}{2}\sum_{k=n+1}^{\infty}\frac{\sigma_{n,k}\sigma_{n,k+2}}{r^{2k+2}}\\
&\leq \frac{r^2+r^{-2}}{4}\sum_{k=0}^{\infty}\frac{\sigma_{n,n+2k+1}^2}{r^{2(n+2k+1)}}\\
&\leq \frac{r^2+r^{-2}}{r^2-r^{-2}}r^{-2n}\,,
\end{aligned}$$

from which the assertion follows. □

Suppose now that $1 < r < R$. Then a function in $f \in H^2(E_R)$ can be identified with a unique function in $H^\infty(E_r)$ by restricting its domain to E_r. The resulting embedding $\mathcal{J} : H^2(E_R) \to H^\infty(E_r)$ is not only continuous, but even compact, by Montel's Theorem. In fact, by the estimates obtained above, it turns out that $\mathcal{J}$ can be efficiently approximated by Legendre polynomials using the following projection operators:

$$\mathcal{P}_n f = \sum_{k=0}^{n-1} a_k(f)P_k \quad (n \in \mathbb{N}_0)\,,$$

with the convention that $\mathcal{P}_0 = 0$.

1.11. Lemma. *Let $1 < r < R$ and let $\mathcal{J} : H^2(E_R) \hookrightarrow H^\infty(E_r)$ denote the canonical embedding. Then*

$$\|\mathcal{J}\|_{H^2(E_R)\hookrightarrow H^\infty(E_r)} \leq \sqrt{\frac{R^2+R^{-2}}{R^2-R^{-2}}}\frac{R^2+Rr}{(R-r)^2}\,.$$

Moreover, for $n \in \mathbb{N}$, we have

$$\|\mathcal{J}-\mathcal{P}_n\|_{H^2(E_R)\hookrightarrow H^\infty(E_r)} \leq (2n+1)\sqrt{\frac{R^2+R^{-2}}{R^2-R^{-2}}}\frac{R^2}{(R-r)^2}\left(\frac{r}{R}\right)^n\,.$$

PROOF. Let $f \in H^2(E_R)$. By a theorem of C. Neumann, see [**1**, Theorem 12.4.7], we can write

$$f(z) = \sum_{k=0}^{\infty} a_k(f) P_k(z)$$

with the series converging uniformly on compact subsets of E_R, hence in particular on E_r. Thus for $n \in \mathbb{N}_0$ we have, using Lemma 1.6 and Lemma 1.10, that

$$\begin{aligned} \|\mathcal{J}f - \mathcal{P}_n f\|_{H^\infty(E_r)} &\leq \sum_{k=n}^{\infty} |a_k(f)| \, \|P_k\|_{H^\infty(E_r)} \\ &\leq \sum_{k=n}^{\infty} (2k+1) \sqrt{\frac{R^2+R^{-2}}{R^2-R^{-2}}} \, \|f\|_{H^2(E_R)} \left(\frac{r}{R}\right)^k . \end{aligned}$$

The assertions now follow by observing that for $z \in \mathbb{C}$ with $|z| < 1$ we have

$$\sum_{k=n}^{\infty} (2k+1) z^k = \frac{(2n+1)z^n - (2n-1)z^{n+1}}{(1-z)^2} \quad (n \in \mathbb{N}_0) .$$

□

We also need the following estimate, which is a variant of the previous lemma.

1.12. LEMMA. *Let $1 < r < R$ and let $\mathcal{J} : H^\infty(E_R) \hookrightarrow H^2(E_r)$ denote the canonical embedding. Then*

$$\|\mathcal{J}\|_{H^\infty(E_R) \hookrightarrow H^2(E_r)} \leq \frac{R + R^{-1}}{R - R^{-1}} \frac{R^2 + Rr}{(R-r)^2} .$$

Moreover, for $n \in \mathbb{N}$, we have

$$\|\mathcal{J} - \mathcal{P}_n\|_{H^\infty(E_R) \hookrightarrow H^2(E_r)} \leq (2n+1) \frac{R + R^{-1}}{R - R^{-1}} \frac{R^2}{(R-r)^2} \left(\frac{r}{R}\right)^n .$$

PROOF. The proof is similar to the proof of the previous lemma, except that we now use Lemma 1.9 instead of Lemma 1.10 together with the trivial bound $|\partial E_R| \leq \pi(R + R^{-1})$. □

References

[1] P.J. Davis. *Interpolation & Approximation*. Dover, New York, 1975.

[2] H. Wang and S. Xiang. On the convergence rates of Legendre polynomials. *Math. Comp.*, 81:861–877, 2012.